



Pleasanton Math Tournament

2026

■ Pleasanton Math Tournament ■ Sprint Round Problems 1–30

HONOR PLEDGE

I pledge to uphold the highest principles of honesty and integrity as a mathlete. I will neither give nor receive unauthorized assistance of any kind. I will not copy another's work and submit it as my own. I understand that any competitor found to be in violation of this honor pledge is subject to disqualification.

Printed Name _____

School _____

DO NOT BEGIN UNTIL YOU ARE INSTRUCTED TO DO SO.

This section of the competition consists of 30 problems. You will have 40 minutes to complete all the problems. You are not allowed to use calculators, books or other aids during this round. If you are wearing a calculator wrist watch, please give it to your proctor now. Calculations may be done on scratch paper. All answers must be complete, legible and simplified to lowest terms. Record only final answers in the blanks in the left-hand column of the competition booklet. If you complete the problems before time is called, use the remaining time to check your answers.

In each written round of the competition, the required unit for the answer is included in the answer blank. The plural form of the unit is always used, even if the answer appears to require the singular form of the unit. The unit provided in the answer blank is the only form of the answer that will be accepted.

Total Correct	Scorer's Initials



Jane
Street



WOLFRAM

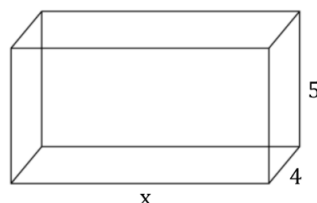
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1. _____ Evaluate $\frac{2026}{2+0+2+6}$. Express your answer as a decimal to the nearest tenth.
2. _____ Arnav notices that adding 76 to his favorite number is equal to tripling his favorite number. What is his favorite number?
3. _____ What is the product of the smallest prime factor of 2025 and the smallest prime factor of 2026?
4. _____ A square has a side length of 3 units. If each side is increased by 1, how much does its area increase by?
5. _____ Vatsal rolls a fair 26 sided dice. What is the probability that he rolls an even prime number?
6. _____ % Makarla attended two meetings during her 84-hour work day. The first meeting took 7 hours and the second meeting took twice as long. What percent of her work day was spent attending meetings?

7. _____ units

The dimensions of a rectangular prism are 4, 5, and x . If the numerical value of the surface area and volume are equal, what is x ?



8. _____ nickels

Tiger has 16 coins totaling 115 cents. If all the coins are either nickels or dimes, how many nickels does Tiger have?

9. _____

Athithan has 10 marbles in a bag: 1 red, 2 yellow, 3 green, and 4 blue. For a certain game he is playing, he will randomly select one marble from the bag, put it back in, and then randomly select another. What is the probability that he selects a red marble, and then a blue marble? Express your answer as a common fraction.

10. _____

What is the largest 4 digit number whose digits multiply to 2025?

11. _____

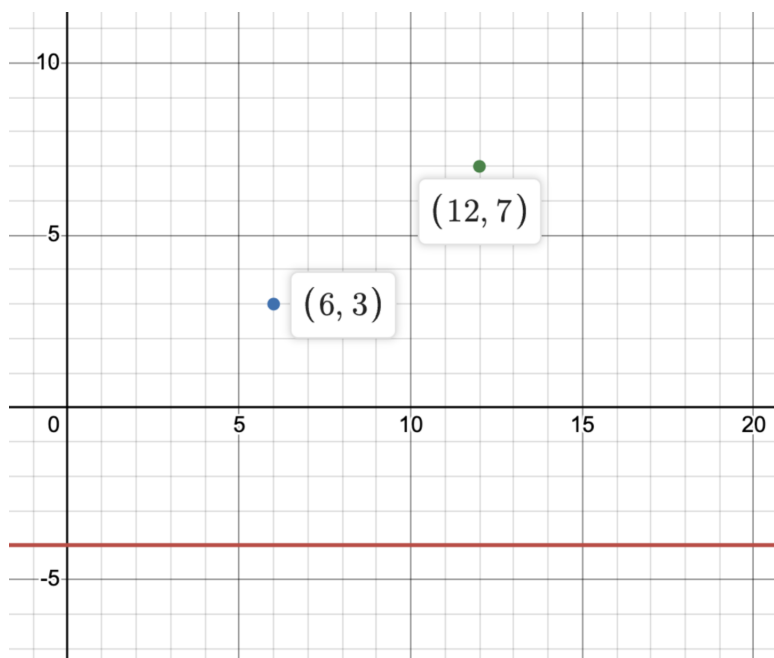
If x and y are integers not necessarily distinct such that $xy = 2025$, what is the positive difference between the maximum and minimum values of $x + y$

12. _____ Pranav has 3 math books, 2 English books, 2 science books, and 1 history book. He wants to arrange them on his book shelf, but keep books of the same subject together. How many ways can he arrange his books if they are all different?
13. _____ cm^2 Isha is running from a bear along a square path. She starts from one corner, runs along the path once, and ends on the same one. If the distance she ran is $4\sqrt{2}$ m, what is the area enclosed by the square path in cm^2 ?
14. _____ If $2025^{P+M+T+2025} = 1$, evaluate $2024^{P+M+T+2024}$
15. _____ Shreyas wrote a number on a paper, but unfortunately he spilled matcha over it. He knows that the number is as large as possible and is divisible by 24. What is the missing digit X if Shreyas could see the the numbers 1234X56?
16. _____ mph Bob travels to work at an acreage of 50 miles per hour. On the way back, there is some traffic and he travels at an average of 30 miles per hour to get back home. He travelled the same distance both ways. What was his average speed in total for the whole trip?
17. _____ How many three-digit numbers contain at least one even digit?

18. _____ What is the remainder when the sum of the coefficients of the expansion of $(x + y)^{2026}$ is divided by 10?
19. _____ Call a number an “3-good” if it has at least 3 positive integer divisors (including one and itself). Call a number “5-good” if it has at least 5 positive integer divisors. How many numbers between 1 and 100 are “3-good” but not “5-good”?
20. _____ Athithan, Vatsal, and Medha are deciding who buys the awards for Pleasanton Math Tournament. If there are 10 identical awards to be won, in how many ways can they split up the purchase if each of them buys at least one award?
21. _____ Five friends Alice, Bob, Charles, Daniel, and Ethan sit in a row. Alice and Bob refuse to sit next to each other, but Charles and Daniel must sit together. How many possible seating arrangements are there?
22. _____ Find the remainder when $2026^{2025^{2024^{\cdot^{2^1}}}}$ is divided by 2025.

23. _____

Tiger lives in the land of the coordinate plane. His mansion is located at $(6, 3)$. A straight river runs along the line $y = -4$. Tiger must stop at some point on the river to fill a bucket and then go to Isha's house at $(12, 7)$ for a party. What is the minimum possible total distance that Tiger can travel?



24. _____

How many 3-digit base 8 integers are also 3 digit integers in both base 7 and base 9?

25. _____

Shreyas has a list of N years which he must memorize for his upcoming AP World History test. He adds every possible pair of these years together. If the last digit of these various sums covers every number from 0 to 9, what is the minimum number of years Shreyas must have on his list?

26. _____

Evaluate $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{2025 \cdot 2026}$.

27. people Shreyas and his 2025 friends organize a Secret Santa. They stand in a circle to exchange gifts, but instead of following a plan, each person randomly gives their gift to one of the two people standing directly next to them, with equal probability. What is the expected number of people who receive 2 presents?
28. points An 11-gon is inscribed inside a circle such that no three of its diagonals intersect at a single point. At how many points do the diagonals intersect strictly inside the circle?
29. Kumar Srivastava the farmer has 4 enclosures for his animals. The first one is shaped like a triangle, with sidelengths 17, 8, and 15. The remaining three are shaped like squares they have one side touching and equal to the corresponding side on the triangle. What is the area of the smallest hexagon which can enclose all four enclosures?
30. The sum of the reciprocals of the altitudes of a triangle is equal to $\frac{2025}{2026}$. What is the inradius of that triangle?